

Beyond Star Ratings: A Context-Aware Restaurant Recommender for NYC

NYU Fundamentals of Machine Learning · Team Noble Jaguars

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FILTERS

Today · 21:45

All NYC

DAY

Today

Tomorrow

Select day

Any time

TIME

Breakfast (9AM)

Lunch (12PM)

Dinner (6PM)

Custom

09:45 PM

WHERE ARE YOU LOOKING?

Select Borough

Search nearby

Select area from map

All NYC

Manhattan

Brooklyn

Queens

Bronx

Staten Island

Searching all 5 boroughs.

Clear

Save

Q e.g. "cozy italian date night outdoor"

Need inspiration?

Browse all restaurants

Search

udon noodles with cozy vibe

Q

X

Results

Share

Raku

4.5

★★★★★

(1,840)

· \$20-30

udon noodles · 48 MacDougal St

Casual Japanese udon and small-plates spot

Closes soon · 9:30 PM · Opens 12 PM

Wed

"Perfect broth - amazing udon - really authentic in a nice atmosphere."

Reserve a table

Order online

Udon St Marks

4.4

★★★★★

(772)

· \$10-20

udon noodles · 11 St Marks Pl

Japanese comfort food

Open · Closes 10 PM

"genuinely best udon ive had in new york and cute date spot tbh"

Order online

←

Clear

Done

Price

\$1.00-\$100.00+

Rating at least

Any rating

Hours

Any time

Open now

Open 24 hours

Dataset Overview

Google Local Reviews — NYC Subset

Source: UCSD Recommender Systems Repository

19,532

Restaurants

≥15 reviews, NYC

2.15M

Review Texts

Full text corpus

4

Boroughs

Full metadata

Rich metadata: categories, hours, price tier (\$-\$\$\$\$), latitude/longitude, ratings

Multilingual reviews: Handled via the dataset's translated field

“

Restaurant Metadata

name | Eleven Madison Park

address | 11 Madison Avenue, New York, NY 10010

avg_rating | 4.8

num_reviews | 2847

price | \$\$\$\$

category | American, Fine Dining

”

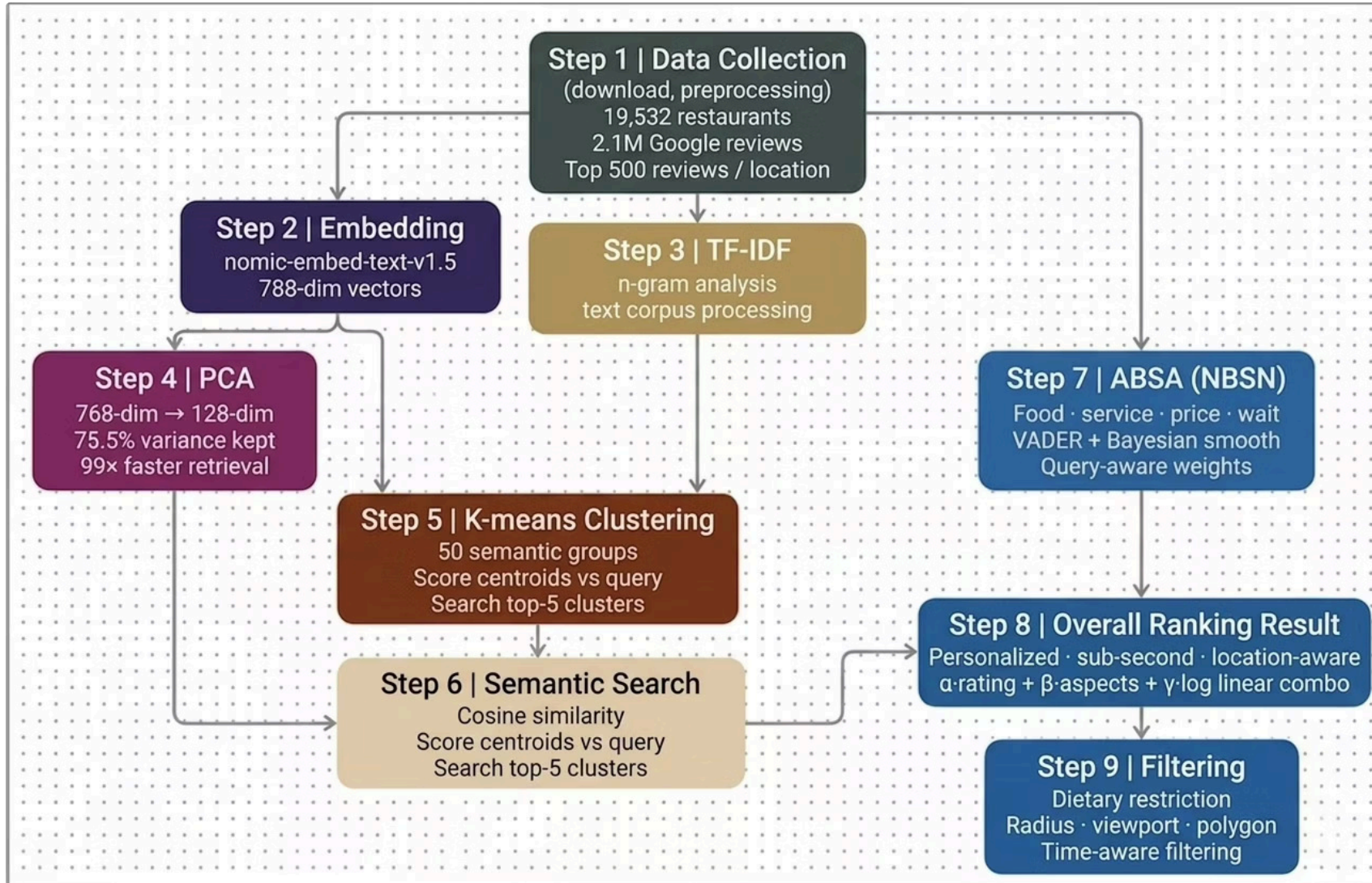
“

Sample Review

Came here for my anniversary. The tasting menu was absolutely exquisite — each course was a work of art. Service was impeccable, staff anticipated our needs before we even asked. The wine pairing was expertly curated. Definitely worth the price tag. Can't wait to go back!

”

System Pipeline



Clustering: Feature Construction

	Meta Embedding (768d)	TF-IDF (500d)
Source	Restaurant metadata (name, category, description)	Aggregated review text per restaurant
Captures	Semantic identity – what it claims to be	Lexical specificity – what reviewers actually say
Method	nomic-embed-text-v1.5	TF-IDF, ngram=(1,2), custom stop words
Example	"Italian restaurant"	"carbonara", "tasting menu", "cash only"

Feature Concatenation

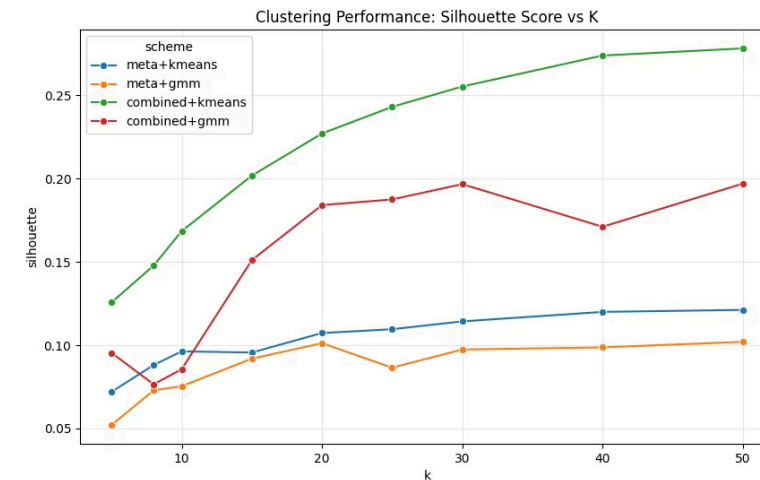
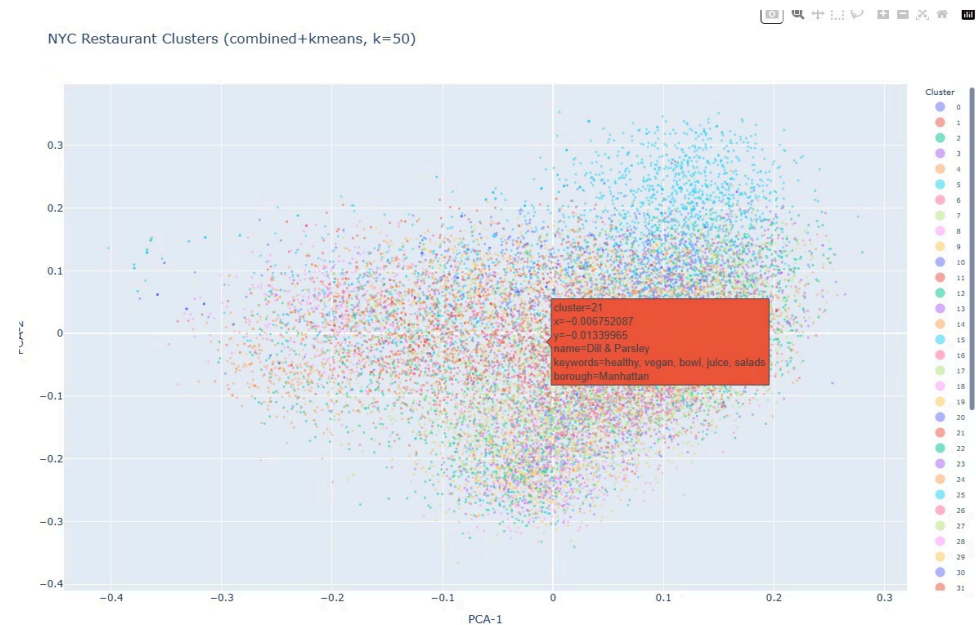
$$\mathbf{x}_{\text{restaurant}} = [\hat{\mathbf{e}}_{\text{meta}} (768d) \parallel \hat{\mathbf{t}}_{\text{tfidf}} (500d)] \in \mathbb{R}^{1268}$$

Both blocks are L2-normalized independently before concatenation

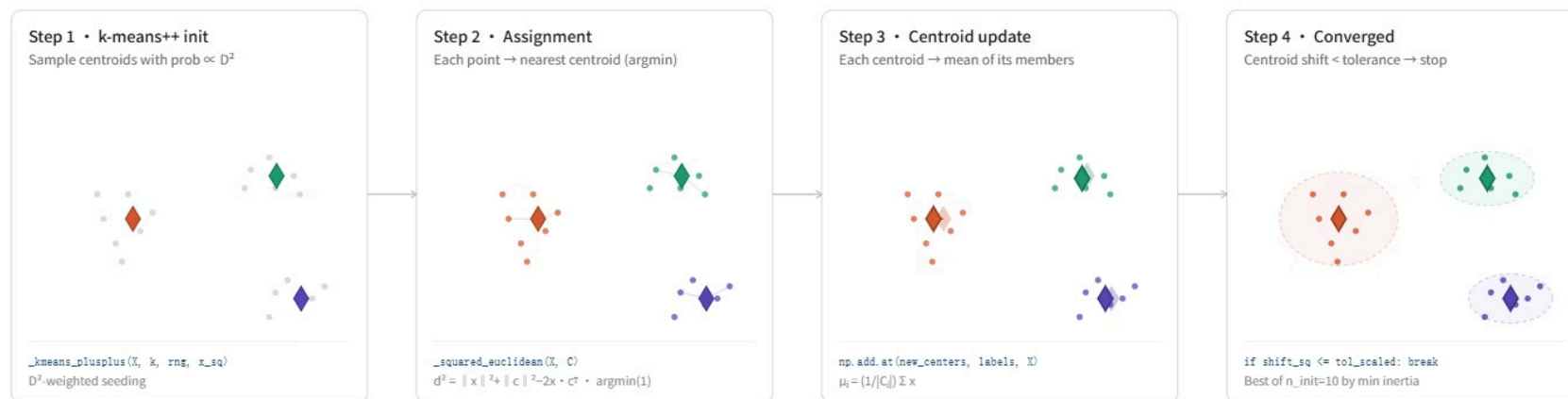


TF-IDF was not covered in lecture. We adopted it from Grootendorst (2022), BERTopic: Neural Topic Modeling with a Class-based TF-IDF Procedure (arXiv:2203.05794).

Cluster Results & K-Means Selection



Combined features beat metadata-only by 138%. K-Means outperforms GMM at every K — supporting the choice of K=50 with combined features.



`class KMeansFromScratch · src/6b_clustering_full.py`

PCA Dimensionality Reduction

We reduced 768-dimensional sentence embeddings to 128 dimensions using Incremental PCA. Achieved 99x speedup (locally) and 83% size reduction while retaining 75.5% of variance. Key insight: recall peaks at 128 and drops for higher counts, suggesting PCA acts as a noise filter.

n_components	Explained Var (GPU)	Recall@10 (GPU)	Speedup (Local)	Speedup (GPU)	Size
16	35.4%	0%	194×	15.6x	132 MB
32	47.2%	6%	160×	8.4x	263 MB
64	60.9%	30%	151×	5.9x	527 MB
128	75.8%	44%	99×	3.2x	1.1 GB
256	89.9%	56%	50×	1.2x	2.1 GB
384	96.2%	66%	34×	0.7x	3.2 GB
512	98.8%	68%	20×	0.2x	4.2 GB

Semantic Search

Our two-stage semantic search pipeline efficiently retrieves relevant restaurants by understanding natural language queries and leveraging pre-clustered data for rapid filtering.

This approach enables a **19x search space reduction**, processing ~113K reviews instead of 2.1M for final ranking and up to 77x for niche queries.

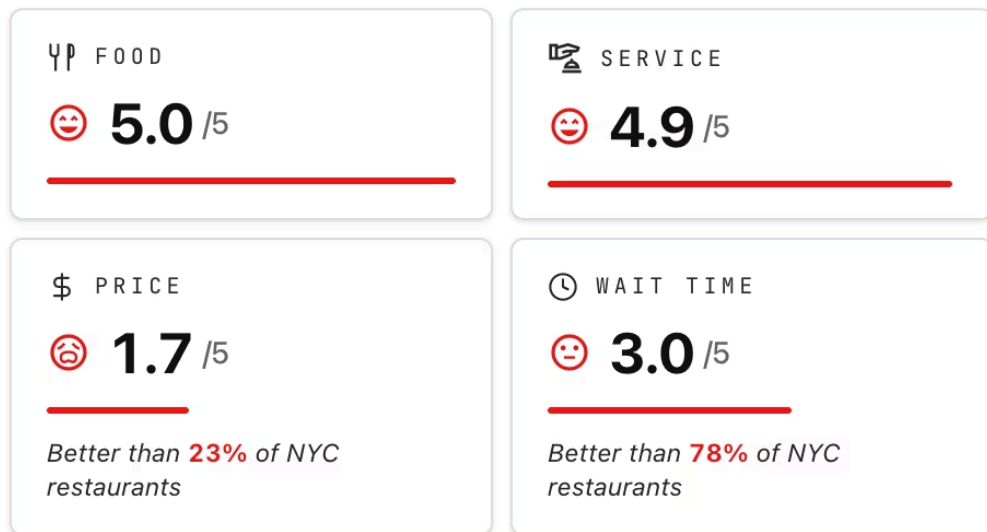
For instance, the query "cheap ramen with fast service" matched clusters 19 (Ramen), 41 (Noodles), and 28 (Curry).

```
<All keys matched successfully>
Searching clusters: [19, 41, 28]
Searching 179171 reviews across 968 restaurants
```

	name	...	gmap_id
2	Terakawa Ramen	...	0x89c25858fc9a1f1f:0x632241386ae392d9
3	Jin Ramen Upper West Side	...	0x89c25885df50bd73:0xb66fad7f5d61101
6	Hinata	...	0x89c258e4f33974e3:0x2c42c9acd1d11845
24	Jun-Men Ramen Bar	...	0x89c259b0ad027815:0x1d22efa7f853e833
11	Tamashii Blue	...	0x89c259238bcca7df:0xe4ab433e2d5e8a7d
19	Rai Rai Ken	...	0x89c2599db4aa5fad:0x151fb895d8fd5263
9	Ramen Takumi	...	0x89c2590671a0e82d:0x8ae2478938bd04e8
29	Ajisen Ramen (Chinatown)	...	0x89c25a26ed785af1:0xf49cc3b69abb0172
22	Ichiran	...	0x89c259af1989db01:0xffff4c602a3069de6
35	Naruto Ramen	...	0x89c2f6260d658d17:0x4b1b4ee080ca77a0

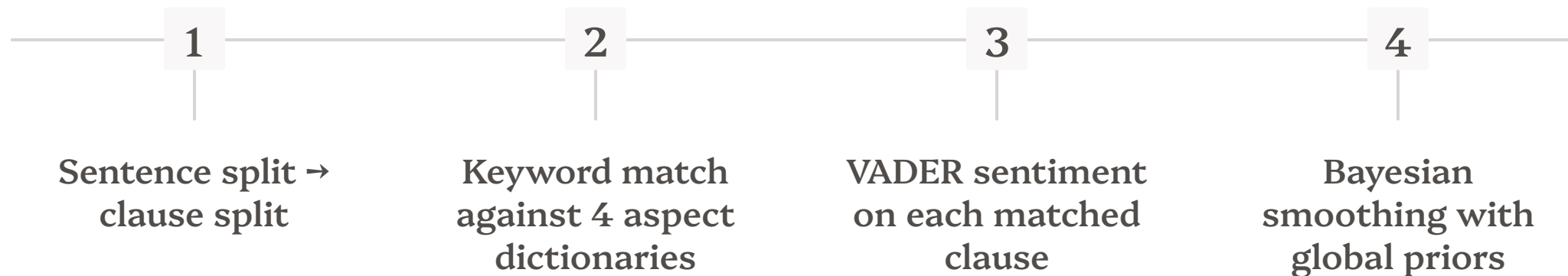
```
[10 rows x 7 columns]
```

ABSA: From Stars to Aspects



- A 4.5-star average doesn't tell you whether it's the food or the service driving that score.
- Our model scores each aspect separately and auto-matches weights to your query.

Technical Pipeline & Ranking



Final ranking formula:

$$\text{aspect_weighted} = w_{\text{food}} \cdot s_{\text{food}} + w_{\text{service}} \cdot s_{\text{service}} + w_{\text{price}} \cdot s_{\text{price}} + w_{\text{wait}} \cdot s_{\text{wait}}$$

Aspect weights are auto-extracted from query language (e.g. "cheap" → $w_{\text{price}} \uparrow$, "quick" → $w_{\text{wait}} \uparrow$).

$$\text{score} = \alpha \cdot \frac{\text{rating}}{5} + \beta \cdot \text{aspect_weighted} + \gamma \cdot \frac{\log(1 + \text{reviews})}{\log(1 + \text{max})}$$

rating $\in [0,5]$ google stars, $\alpha=0.4$, $\beta=0.5$, $\gamma=0.1$ (tuned via sensitivity analysis)

Review count serves as a credibility signal, log-scaled captures this with diminishing